

Conforming lifting and adjoint consistency for the Discrete de Rham complex of differential forms

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joint work with D. Di Pietro and J. Droniou

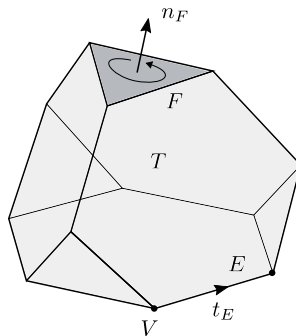
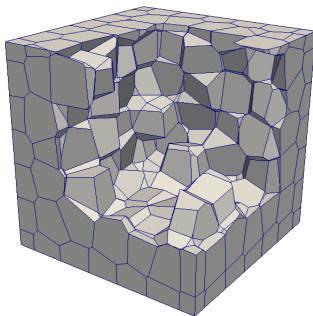
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The Discrete de Rham construction

- 1 The Discrete de Rham construction
- 2 Objective and motivation
- 3 Construction of the Lifting
- 4 Solving the two abstract problems
- 5 Conclusions

Domain and polytopal mesh

- $\Omega \subset \mathbb{R}^n$ polyhedral domain
- $\mathcal{M}_h$ polyhedral mesh on Ω
- $\Delta_d(\mathcal{M}_h)$ set of d -cells



Discrete de Rham method with differential forms

- The de Rham complex with differential forms:

$$H\Lambda^0(\Omega) \xrightarrow{d^0} \dots \xrightarrow{d^{k-1}} H\Lambda^k(\Omega) \xrightarrow{d^k} \dots \xrightarrow{d^{n-1}} H\Lambda^n(\Omega)$$

- The **DDR complex** of **polynomial differential forms** ⁽¹⁾:

$$\underline{X}_{r,h}^0 \xrightarrow{\underline{d}_{r,h}^0} \dots \xrightarrow{\underline{d}_{r,h}^{k-1}} \underline{X}_{r,h}^k \xrightarrow{\underline{d}_{r,h}^k} \dots \xrightarrow{\underline{d}_{r,h}^{n-1}} \underline{X}_{r,h}^n$$

Design principle: recursive construction on d -cells (e.g., k -form on $d = k + 1, \dots, n$) mimicking the Stokes formula

¹[Bonaldi et al., 2023]

Trimmed polynomial spaces

- Let $f \in \Delta_d(\mathcal{M}_h)$, fix $\mathbf{x}_f \in f$
- **Koszul complement:**

$$\mathcal{K}_r^\ell(f) := \kappa \mathcal{P}_{r-1} \Lambda^{\ell+1}(f)$$

with $(\kappa\omega)_{\mathbf{x}}(\cdot, \dots) := \omega_{\mathbf{x}}(\mathbf{x} - \mathbf{x}_f, \dots)$

Trimmed polynomial spaces

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- **Trimmed polynomial spaces** ($\ell \geq 1$):

$$\mathcal{P}_r^- \Lambda^0(f) := \mathcal{P}_r \Lambda^0(f),$$

$$\mathcal{P}_r^- \Lambda^\ell(f) := d\mathcal{P}_r \Lambda^{\ell-1}(f) \oplus \mathcal{K}_r^\ell(f)$$

Remarks:

- *Full space:* $\mathcal{P}_r \Lambda^\ell(f) = d\mathcal{P}_{r+1} \Lambda^{\ell-1}(f) \oplus \mathcal{K}_r^\ell(f)$

Discrete local potential and exterior derivative (I)

- Discrete potential (playing the role of a k -form inside f)

$$P_{r,f}^k : \underline{X}_{r,f}^k \rightarrow \mathcal{P}_r \Lambda^k(f)$$

- Discrete exterior derivative

$$d_{r,f}^k : \underline{X}_{r,f}^k \rightarrow \mathcal{P}_r \Lambda^{k+1}(f)$$

Discrete local potential and exterior derivative (I)

For $d = k, \dots, n$, all $f \in \Delta_d(\mathcal{M}_h)$, and $\underline{\omega}_f \in \underline{X}_{r,f}^k$:

- **Case** $d = k$:

$$P_{r,f}^k \underline{\omega}_f := \star^{-1} \omega_f \in \mathcal{P}_r \Lambda^d(f)$$

- **Case** $d \geq k + 1$:

- Discrete exterior derivative $d_{r,f}^k \underline{\omega}_f \in \mathcal{P}_r \Lambda^{k+1}(f)$ defined such that,
 $\forall \mu \in \mathcal{P}_r \Lambda^{d-k-1}(f)$:

$$\int_f d_{r,f}^k \underline{\omega}_f \wedge \mu = (-1)^{k+1} \int_f \star^{-1} \omega_f \wedge d\mu + \int_{\partial f} P_{r,\partial f}^k \underline{\omega}_{\partial f} \wedge \text{tr}_{\partial f} \mu$$

Discrete local potential and exterior derivative (II)

Case $d \geq k + 1$ (continued):

- **Discrete potential** $P_{r,f}^k \underline{\omega}_f \in \mathcal{P}_r \Lambda^k(f)$ defined using the decomposition $\mathcal{P}_r \Lambda^{d-k}(f) = d\mathcal{K}_{r+1}^{d-k-1}(f) \oplus \mathcal{K}_r^{d-k}(f)$:

- For all $\mu \in \mathcal{K}_{r+1}^{d-k-1}(f)$:

$$(-1)^{k+1} \int_f P_{r,f}^k \underline{\omega}_f \wedge d\mu = \int_f d_{r,f}^k \underline{\omega}_f \wedge \mu - \int_{\partial f} P_{r,\partial f}^k \underline{\omega}_{\partial f} \wedge \text{tr}_{\partial f} \mu$$

- For all $\nu \in \mathcal{K}_r^{d-k}(f)$:

$$\int_f \star^{-1} P_{r,f}^k \underline{\omega}_f \wedge \nu = \int_f \star^{-1} \omega_f \wedge \nu$$

Discrete local potential and exterior derivative (III)

Lemma (Polynomial consistency)

For all integers $0 \leq k \leq d \leq n$ and all $f \in \Delta_d(\mathcal{M}_h)$, it holds

$$P_{r,f}^k I_{r,f}^k \omega = \omega \quad \forall \omega \in \mathcal{P}_r \Lambda^k(f),$$

and, if $d \geq k + 1$,

$$d_{r,f}^k I_{r,f}^k \omega = d\omega \quad \forall \omega \in \mathcal{P}_{r+1}^- \Lambda^k(f).$$

Discrete $X_{r,h}^k$ spaces : hierarchical structure

$$\underline{X}_{r,h}^k := \bigtimes_{d=k}^n \bigtimes_{f \in \Delta_d(\mathcal{M}_h)} \star^{-1} \mathcal{P}_r^- \Lambda^{d-k}(f)$$

For $\omega \in \Lambda^k(\Omega)$, $\underline{I}_{r,h}^k \omega = (\pi_{r,f}^{-,d-k}(\star \text{tr}_f \omega))_{f \in \Delta_k(\mathcal{M}_h)} \in \underline{X}_{r,h}^k$

Space	$f_0 \equiv V$	$f_1 \equiv E$	$f_2 \equiv F$	$f_3 \equiv T$
$\underline{X}_{r,h}^0$	$\mathbb{R} = \star^{-1} \mathcal{P}_r \Lambda^0(f_0)$	$\star^{-1} \mathcal{P}_r^- \Lambda^1(f_1)$	$\star^{-1} \mathcal{P}_r^- \Lambda^2(f_2)$	$\dots$
$\underline{X}_{r,h}^1$		$\star^{-1} \mathcal{P}_r \Lambda^0(f_1)$	$\star^{-1} \mathcal{P}_r^- \Lambda^1(f_2)$	$\dots$
$\underline{X}_{r,h}^2$			$\star^{-1} \mathcal{P}_r \Lambda^0(f_2)$	$\dots$
$\underline{X}_{r,h}^3$				$\dots$

Discrete $X_{r,h}^k$ spaces : L^2 -products

- We can define on $X_{r,h}^k$ a **discrete L^2 -product**:

$$(\underline{\omega}_h, \underline{\mu}_h)_{k,h} := \sum_{f \in \Delta_n(\mathcal{M}_h)} \left(\int_f P_{r,f}^k \underline{\omega}_f \wedge \star P_{r,f}^k \underline{\mu}_f + s_{k,f}(\underline{\omega}_f, \underline{\mu}_f) \right)$$

- Above, $s_{k,f}$ is a **stabilisation term** such that

$$s_{k,f}(I_{r,f}^k \omega, \underline{\mu}_f) = 0 \quad \forall \omega \in \mathcal{P}_r \Lambda^k(f)$$

Global discrete exterior derivative and DDR complex

- Global discrete exterior derivative: $\underline{d}_{r,h}^k : \underline{X}_{r,h}^k \rightarrow \underline{X}_{r,h}^{k+1}$

$$\underline{d}_{r,h}^k \underline{\omega}_h := \left(\pi_{r,f}^{-,d-k-1} (\star \underline{d}_{r,f}^k \underline{\omega}_f) \right)_{f \in \Delta_{[k+1 \dots n]}(\mathcal{M}_h)}$$

- DDR sequence:

$$\underline{X}_{r,h}^0 \xrightarrow{\underline{d}_{r,h}^0} \dots \xrightarrow{\underline{d}_{r,h}^{k-1}} \underline{X}_{r,h}^k \xrightarrow{\underline{d}_{r,h}^k} \dots \xrightarrow{\underline{d}_{r,h}^{n-1}} \underline{X}_{r,h}^n$$

Global discrete exterior derivative and DDR complex

- DDR sequence:

$$\underline{X}_{r,h}^0 \xrightarrow{\underline{d}_{r,h}^0} \cdots \xrightarrow{\underline{d}_{r,h}^{k-1}} \underline{X}_{r,h}^k \xrightarrow{\underline{d}_{r,h}^k} \cdots \xrightarrow{\underline{d}_{r,h}^{n-1}} \underline{X}_{r,h}^n$$

Lemma (Cohomology of the Discrete de Rham complex)

The DDR sequence is a complex and its cohomology is isomorphic to the cohomology of the continuous de Rham complex, i.e., for all k ,

$$\text{Ker } \underline{d}_{r,h}^k / \text{Im } \underline{d}_{r,h}^{k-1} \cong \text{Ker } d^k / \text{Im } d^{k-1}.$$

Global discrete exterior derivative and DDR complex

- DDR sequence:

$$\underline{X}_{r,h}^0 \xrightarrow{\underline{d}_{r,h}^0} \dots \xrightarrow{\underline{d}_{r,h}^{k-1}} \underline{X}_{r,h}^k \xrightarrow{\underline{d}_{r,h}^k} \dots \xrightarrow{\underline{d}_{r,h}^{n-1}} \underline{X}_{r,h}^n$$

Lemma (Poincaré inequalities)

For any $k = 0, \dots, n-1$ and $\underline{\omega}_h \in \underline{X}_{r,h}^k$, there exists $\underline{\tau}_h \in \underline{X}_{r,h}^k$ such that $\underline{d}_{r,h}^k \underline{\tau}_h = \underline{d}_{r,h}^k \underline{\omega}_h$ and

$$\|\underline{\tau}_h\|_h \lesssim \|\underline{d}_{r,h}^k \underline{\omega}_h\|_h.$$

See presentation by **Marien Hanot**

Objective and motivation

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Goal: conforming lifting operator

Build a linear operator

$$\mathcal{L}_{r,h}^k : \underline{X}_{r,h}^k \rightarrow H\Lambda^k(\Omega)$$

Key Properties: for each $d \in \{k, \dots, n\}$ and $f \in \Delta_d(\mathcal{M}_h)$,

- **Projection:** for all $\zeta \in \mathcal{P}_{r+1}^- \Lambda^{d-k}(f)$,

$$\int_f \mathrm{tr}_f(\mathcal{L}_{r,h}^k \underline{\omega}_h) \wedge \zeta = \int_f P_{r,f}^k \underline{\omega}_f \wedge \zeta$$

- **Stability:**

$$\|\mathrm{tr}_f(\mathcal{L}_{r,h}^k \underline{\omega}_h)\|_f \lesssim \|\underline{\omega}_f\|_f$$

$$\|\mathrm{d} \mathrm{tr}_f(\mathcal{L}_{r,h}^k \underline{\omega}_h)\|_f \lesssim \|\mathrm{d}_{r,f}^k \underline{\omega}_f\|_f$$

Motivation : proof of adjoint consistency (I)

Adjoint consistency: $\alpha \in H\Lambda^k(\Omega)$ continuous k -form, $\underline{\omega}_h \in \underline{X}_{r,h}^{n-k-1}$,

$$\left| \underbrace{\int_{\Omega} \alpha \wedge \underline{d}_{r,h}^{n-k-1} \underline{\omega}_h - (-1)^{k+1} \int_{\Omega} d\alpha \wedge P_{r,h}^{n-k-1} \underline{\omega}_h}_{\mathfrak{J}} \right| \lesssim h^{r+1} \mathcal{N}(\alpha, \underline{\omega}_h)$$

Motivation : proof of adjoint consistency (II)

Proof steps:

- ① Introduce local polynomial approximations $(\mu_f)_{f \in \mathcal{T}_h}$ of α :

$$\begin{aligned} \mathfrak{I} = & \sum_{f \in \Delta_n(\mathcal{M}_h)} \left(\int_f (\alpha - \mu_f) \wedge \underline{d}_{r,f}^{n-k-1} \underline{\omega}_f \right. \\ & + (-1)^{k+1} \int_f d(\mu_f - \alpha) \wedge P_{r,f}^{n-k-1} \underline{\omega}_f \\ & \left. + (-1)^k \sum_{f' \in \Delta_{n-1}(f)} \epsilon_{ff'} \int_{f'} \text{tr}_{f'}(\mu_f) \wedge P_{r,f'}^{n-k-1} \underline{\omega}_{f'} \right) \end{aligned}$$

- ② Introduce $\mathcal{L}_{r,h}^{n-k-1}$, replace potential P , with lifting trace:

$$P_{r,f'}^{n-k-1} \underline{\omega}_{f'} \longrightarrow \text{tr}_{f'}(\mathcal{L}_{r,h}^{n-k-1} \underline{\omega}_h)$$

- ③ Apply **Integration By Parts** on the continuous lifting to close the estimate

Motivation : stabilization free DDR methods

- **Principle:** the bilinear form is built by evaluating the continuous operator directly on the lifting $\mathcal{L}_{r,h}^k$
- **No stabilization term:** coercivity is assured by the piecewise definition and properties of $\mathcal{L}_{r,h}^k$

Example: the discrete bilinear form for the Hodge-Laplace problem reads

$$a_h(\omega_h, \nu_h) := \int_{\Omega} d(\mathcal{L}_{r,h}^k \omega_h) \wedge \star d(\mathcal{L}_{r,h}^k \nu_h)$$

Construction of the Lifting

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Principle behind the construction

Recursive construction: on cell dimension d , from k to n

- **Base case** $d = k$:

$$\text{Define } \mathcal{L}_{r,f}^k \omega_f := P_{r,f}^k \omega_f$$

- **Inductive step** $d > k$:

- Find $\lambda \in H\Lambda^k(f)$ such that

$$\begin{cases} d\lambda = d_{r,f}^k \omega_f + \mathcal{C}_{r,f}^k \omega_f \\ \text{tr}_{\partial f} \lambda = \mathcal{L}_{r,\partial f}^k \omega_{\partial f} \end{cases}$$

with $\mathcal{C}_{r,f}^k$ a **lifting correction** (non-zero only if $d \geq k + 2$)

- Find $\tau \in H\Lambda_0^{k-1}(f)$ such that

$$\int_f d\tau \wedge \nu = \int_f (P_{r,f}^k \omega_f - \lambda) \wedge \nu, \quad \forall \nu \in \mathcal{K}_r^{d-k}(f)$$

$$\text{Define } \mathcal{L}_{r,f}^k \omega_f := \lambda + d\tau$$

Verification of properties: Projection

Goal: Prove the **projection property** for all $\zeta \in \mathcal{P}_{r+1}^- \Lambda^{d-k}(f)$

$$\int_f \mathrm{tr}_f(\mathcal{L}_{r,h}^k \underline{\omega}_h) \wedge \zeta = \int_f P_{r,f}^k \underline{\omega}_f \wedge \zeta$$

Proof sketch: Use decomposition $\zeta = d\mu + \nu$

- **Step 1:** For $d\mu$ (with $\mu \in \mathcal{K}_r^{d-k}(f)$):

$$\begin{aligned} \int_f \mathcal{L}_{r,f}^k \underline{\omega}_f \wedge d\mu &= (-1)^{k+1} \int_f d\mathcal{L}_{r,f}^k \underline{\omega}_f \wedge \mu + \text{boundary terms} \\ &\stackrel{(*)}{=} (-1)^{k+1} \int_f d_{r,f}^k \underline{\omega}_f \wedge \mu + \dots \\ &= \int_f P_{r,f}^k \underline{\omega}_f \wedge d\mu \end{aligned}$$

(*) using $d\mathcal{L}_{r,f}^k = d_{r,f}^k + \mathcal{C}_{r,f}^k$

Verification of properties: Projection

Goal: Prove the **projection property** for all $\zeta \in \mathcal{P}_{r+1}^- \Lambda^{d-k}(f)$

$$\int_f \mathrm{tr}_f(\mathcal{L}_{r,h}^k \underline{\omega}_h) \wedge \zeta = \int_f P_{r,f}^k \underline{\omega}_f \wedge \zeta$$

Proof sketch: Use decomposition $\zeta = d\mu + \nu$

- **Step 2:** For $\nu \in \mathcal{K}_r^{d-k+1}(f)$:

$$\int_f (\lambda + d\tau) \wedge \nu = \int_f P_{r,f}^k \underline{\omega}_f \wedge \nu$$

Guaranteed by the construction of τ .

Verification of properties: Stability

Goal: Prove the bound $\|\mathrm{tr}_f(\mathcal{L}_{r,h}^k \underline{\omega}_h)\|_f \lesssim \|\underline{\omega}_f\|_f$

Proof idea: The lifting is defined as $\mathcal{L}_{r,f}^k \underline{\omega}_f = \lambda + d\tau$

- ① λ is the **minimal norm solution** to a potential problem $\Rightarrow$ its norm is controlled by the data (RHS and boundary conditions)

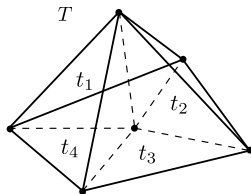
$$\|\lambda\|_f \lesssim \|d_{r,f}^k \underline{\omega}_f\|_f + \|\mathcal{L}_{r,\partial f}^k \underline{\omega}_{\partial f}\|_{\partial f}$$

- ② $d\tau$ has zero trace on ∂f and is constructed to match the projection $\Rightarrow$

$$\|d\tau\|_f \lesssim \|P_{r,f}^k \underline{\omega}_f - \lambda\|_f$$

A preliminary step: FEEC spaces on a simplicial submesh

Simplicial submesh: simplicial subdivision $\mathcal{S}_h(f)$ of each cell $f \in \Delta_d(\mathcal{M}_h)$



FEEC spaces on $\mathcal{S}_h(f)$:

$$\dots \longrightarrow \mathcal{P}_r^- \Lambda^k(\mathcal{S}_h(f)) \xrightarrow{\text{d}} \mathcal{P}_r^- \Lambda^{k+1}(\mathcal{S}_h(f)) \longrightarrow \dots$$

Towards two abstract problems

Let $f \in \Delta_d(\mathcal{M}_h)$

Data: $\xi \in \mathcal{P}_r^- \Lambda^{k+1}(\mathcal{S}_h(f))$, $\theta \in \mathcal{P}_{r+1}^- \Lambda^k(\mathcal{S}_h(\partial f))$, $\zeta \in L^2 \Lambda^k(f)$

Find:

① $\lambda \in \mathcal{P}_{r+1}^- \Lambda^k(\mathcal{S}_h(f))$ such that

$$\begin{cases} d\lambda = \xi \\ \text{tr}_{\partial f} \lambda = \theta \end{cases}$$

② $\tau \in \mathring{\mathcal{P}}_r^- \Lambda^{k-1}(\mathcal{S}_h(f))$ such that, for all d -simplex $t \in \mathcal{S}_h(f)$,

$$-\int_t \tau \wedge dz = \int_t \zeta \wedge z, \quad \forall z \in \mathcal{K}_r^{d-k}(t)$$

Towards two abstract problems

Let $f \in \Delta_d(\mathcal{M}_h)$

Data: $\xi \in \mathcal{P}_r^- \Lambda^{k+1}(\mathcal{S}_h(f))$, $\theta \in \mathcal{P}_{r+1}^- \Lambda^k(\mathcal{S}_h(\partial f))$, $\zeta \in L^2 \Lambda^k(f)$

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$$-\int_t \tau \wedge dz = \int_t \zeta \wedge z, \quad \forall z \in \mathcal{K}_r^{d-k}(t)$$

Novel aspects of the construction

- **Simplicial submesh + Dropping co-differential constraint** (i.e., not solving **curl**-div systems) $\Rightarrow$ no need to assume solution regularity
- **Explicit polynomial expression** $\Rightarrow$ convenient to work with and allows for straightforward stability estimates

Solving the two abstract problems

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Reminder : two abstract problems

Let $f \in \Delta_d(\mathcal{M}_h)$,

Data: $\xi \in \mathcal{P}_r^- \Lambda^{k+1}(\mathcal{S}_h(f))$, $\theta \in \mathcal{P}_{r+1}^- \Lambda^k(\mathcal{S}_h(\partial f))$, $\zeta \in L^2 \Lambda^k(f)$

Find:

① $\lambda \in \mathcal{P}_{r+1}^- \Lambda^k(\mathcal{S}_h(f))$ such that

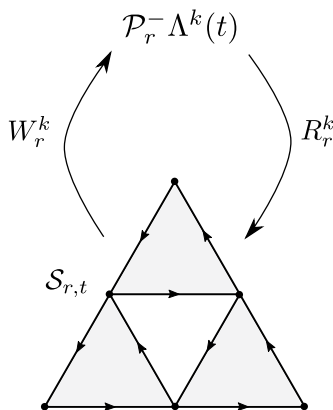
$$\begin{cases} d\lambda = \xi \\ \text{tr}_{\partial f} \lambda = \theta \end{cases}$$

② $\tau \in \mathring{\mathcal{P}}_?^- \Lambda^{k-1}(\mathcal{S}_h(f))$ such that, for all d -simplex $t \in \mathcal{S}_h(f)$,

$$-\int_t \tau \wedge dz = \int_t \zeta \wedge z, \quad \forall z \in \mathcal{K}_k^\ell(t)$$

First problem: solving via small simplices

Let $t \in \Delta_d(\mathcal{S}_h(f))$



Cochain spaces: $C^k(\mathcal{S}_{r,t}) \cong \mathbb{R}^{\text{card}(\Delta_k(\mathcal{S}_{r,t}))}$

DoFs on small simplices

DoFs on small simplices

Define

$$C^k(\mathcal{S}_{r,t}, \mathcal{H}_{r,t}) := \text{Im } R_r^k.$$

Then the graded spaces $C^k(\mathcal{S}_{r,t}, \mathcal{H}_{r,t})$, together with the differential δ^k (**coboundary operator**), form a cochain subcomplex of $C^k(\mathcal{S}_{r,t})$.

Moreover, there exists a **cochain isomorphism**

$$\varphi_r^k : C^k(\mathcal{S}_{r,t}, \mathcal{H}_{r,t}) \rightarrow \mathcal{P}_r^- \Lambda^k(t)$$

An equivalent problem on small simplices

For $f \in \Delta_d(\mathcal{M}_h)$

Data: $\tilde{\xi} \in C^{k+1}(\mathcal{S}_{r,f}, \mathcal{H}_{r,f})$ and $\tilde{\theta} \in C^k(\mathcal{S}_{r,f}, \mathcal{H}_{r,f})$

Find: $\tilde{\lambda} \in C^k(\mathcal{S}_{r,f}, \mathcal{H}_{r,f})$ such that

$$\begin{cases} \delta^k \tilde{\lambda} = \tilde{\xi}, \\ \gamma_{\partial f}^k \tilde{\lambda} = \tilde{\theta} \end{cases}$$

Existence is proven by an explicit construction of a solution !

Reminder : two abstract problems

Let $f \in \Delta_d(\mathcal{M}_h)$

Data: $\xi \in \mathcal{P}_r^- \Lambda^{k+1}(\mathcal{S}_h(f))$, $\theta \in \mathcal{P}_{r+1}^- \Lambda^k(\mathcal{S}_h(\partial f))$, $\zeta \in L^2 \Lambda^k(f)$

Find:

① $\lambda \in \mathcal{P}_{r+1}^- \Lambda^k(\mathcal{S}_h(f))$ such that

$$\begin{cases} d\lambda = \xi \\ \text{tr}_{\partial f} \lambda = \theta \end{cases}$$

② $\tau \in \mathring{\mathcal{P}}_r^- \Lambda^{k-1}(\mathcal{S}_h(f))$ such that, for all d -simplex $t \in \mathcal{S}_h(f)$,

$$-\int_t \tau \wedge dz = \int_t \zeta \wedge z, \quad \forall z \in \mathcal{K}_r^{d-k}(t)$$

Second problem

Hodge duality on FEEC spaces

Let t be a d -simplex. Then, for any nonzero $\alpha \in \mathcal{P}_r \Lambda^k(t)$ there exists a form $\beta \in \mathring{\mathcal{P}}_{r+d-k+1}^- \Lambda^{d-k}(t)$ such that

$$\int_t \beta \wedge \alpha > 0$$

Corollary: the bilinear form $\int_t(\cdot, \cdot)$ is non-degenerate, and for the linear functional

$$L(dz) = \int_t \zeta \wedge z, \quad z \in \mathcal{K}_r^{d-k}(t)$$

there exists a unique $\tau \in \mathring{\mathcal{P}}_{r+d-k+1}^- \Lambda^{k-1}(t)$ such that

$$\int_t \tau \wedge dz = L(dz).$$

Conclusions

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Take-home message

- **General strategy:** Solving underdetermined potential problems via FEEC spaces on simplicial submeshes
- **Corollaries:**
 - A framework to establish analytical results on polytopal complexes (e.g., Poincaré inequalities, adjoint consistency, compactness)
 - **Explicit and computable** conforming representations of discrete data (enabling stabilization-free FEM on polyhedral meshes)

Thank you for your attention!



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Selected Publications

 Di Pietro, D. A., Droniou, J., & Pitassi, S. (2025).

Conforming lifting and adjoint consistency for the Discrete de Rham complex of differential forms.

arXiv preprint

 Alonso Rodríguez, A., Pitassi, S., & Rapetti, F.

Poincaré-constants estimates on Finite Element simplicial stars: construction via high-order potentials.

In preparation

References I



Bonaldi, F., Di Pietro, D. A., Droniou, J., and Hu, K. (2023).

An exterior calculus framework for polytopal methods.

<http://arxiv.org/abs/2303.11093>.